Tank Spotter App — Design Overview

A spec document summarizing design decisions for a gamified tank identification app built on spaced-repetition (Anki-like) principles.

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1. Core Concept

A tank identification game where players are shown an image and must pick the correct tank from 4 options. Cards seen frequently appear less often (spaced repetition). The app should be **broad and accessible** — supporting casual learners, history enthusiasts, and OSINT-curious users alike — and must work across all session lengths (30-second bursts to 15+ minute sessions).

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2. The Two Foundational Problems

2.1 The "one image per tank" problem

Currently each of the ~400 tanks has a single image. This means spaced repetition rehearses **pixels, not tank features**. Once a player has seen the deck, the game becomes trivial. **This is the primary content bottleneck.**

2.2 The "97% accuracy" problem

Many players currently sit at ~97% accuracy because the easy mode is too easy. Target: only **~1% of players above 90%** accuracy. Accuracy must become a meaningful skill signal, not an inflated metric.

Both problems are solved by the same mechanism: **procedural transforms feeding an adaptive difficulty system.**

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3. Procedural Transforms (Content Multiplier)

One source image yields 6–10 distinct cards via transforms. Each transform also functions as a **difficulty tier** within gameplay.

Transform	Tier Name	Difficulty
Original photo	Standard	1x
Horizontal flip	Flipped	1.2x
Aggressive crop (turret / hull / running gear)	Detail	1.5x
Black-fill silhouette	Silhouette	2x
Edge-detection / thermal LUT	Thermal Recon	2x
Heavy desaturation + grain	Archival	1.8x
Partial occlusion (foliage/smoke overlay)	Obscured	1.8x
Low-res pixelation	Satellite Intel	2.2x
Trick variations (4 options = variants of same tank)	Trick	2x
Type-the-name (no options shown)	Identification	3x

Build at runtime, not build-time. Apply transforms client-side from a single source image — smaller asset bundle, more flexible, can add new transforms without re-shipping.

War Thunder screenshots are an acceptable supplementary source — same model, multiple angles, varied lighting and camo.

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4. Adaptive Difficulty System

Track **rolling accuracy over last 50 cards** (not all-time — stale data shouldn't lock difficulty). The card type distribution served to a player is a function of this rolling number.

Rolling Accuracy	Standard	Flipped/Cropped	Trick	Silhouette/Thermal	Type-name
<60%	95%	5%	—	—	—
60–75%	75%	20%	5%	—	—
75–85%	50%	30%	15%	5%	—
85–92%	25%	30%	25%	15%	5%
92%+	10%	25%	25%	25%	15%

This is **self-balancing**: the player at 97% gets brutal cards and drifts to ~88%. The player at 60% gets eased. Everyone converges to their true skill level.

Migration warning: when this ships, currently-inflated users will see their accuracy drop. Mitigate by showing a separate **Spotter Rating (ELO, monotonically increasing)** alongside accuracy — messaging is "your skill rating reflects how well you handle harder content."

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5. Scoring Formula

Per-card points are **multiplicative**, not additive:

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Points = CardDifficulty × TimeBonus × StreakMultiplier × ModeMultiplier
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CardDifficulty (dynamic)

Computed from **% of players globally who get this card right**. A T-90M that 12% of players ID correctly is worth far more than a Sherman everyone knows. This makes rare tanks feel rewarding to identify and prevents grinding on easy cards.

TimeBonus

- 2-second "reading buffer" — no decay
- Linear decay from t=2s to t=8s
- Floor at 50% (never zero — kills rage-clicking incentive)

StreakMultiplier

- Caps at 2x at 10-streak (so casuals can reach it)
- Streak breaks on any wrong answer

ModeMultiplier

See table in Section 3. Standard = 1x, transforms = 1.2–2.2x, type-the-name = 3x.

Confidence Betting

Before the 4 options appear, player can tap a "Confident" button to wager 2x.

- Correct → double points
- Wrong → double loss

Rewards genuine knowledge over reflex guessing and pairs especially well with trick-variation tier.

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6. Linear Rank Progression

XP-driven, monotonic — always ticks up. Time-based, fair to time-poor players. Used for long-term progression.

Proposed thematic ladder (intelligence/recon, not infantry):
Observer → Scout → Recon Specialist → Intelligence Analyst → Senior Analyst → Field Officer → Recon Commander → Intelligence Director → Spymaster

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7. Build Priority Order

1. **Procedural transforms pipeline** on existing 400 tanks (~6x content multiplier, ships first)
2. **Adaptive difficulty mix** based on rolling 50-card accuracy (fixes the 97% problem)
3. **Spotter Rating (ELO)** + dynamic card difficulty weighting (fair scoring across skill levels)
4. **Linear rank progression** (long-term goal for players)

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8. Open Design Decisions

Things still to decide:

- Exact XP curve for linear rank progression
- ELO K-factor for Spotter Rating
- Naming and visual treatment of the rank ladder
- Free-text vs structured input for Type-the-Name tier (fuzzy matching vs blanked template)
- Whether linear rank gates transform tiers, or adaptive difficulty handles tier exposure alone

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9. Non-Goals (Explicit)

- **No monetization in v1.** Possibly future, not the goal now.
- **No paid-only mechanics.** Every "scarce" resource must be earnable through play.
- **No leaderboards in v1.** Focus on a game that works first.
- **No reliance on one image per tank** — the procedural transforms pipeline must ship before any other gameplay work.